



ANAND ANATHUR

ELANGOVAN

ENGR III CSLT-SYS ARCHITECT

PROFESSIONAL PROFILE

CONTACT



+91 8778585263



anandanathurelangovan94@gmail.com



www.linkedin.com/in/anand-anathur-elangovan-50339b419



Chennai, TN, India

TECHNICAL SKILLS

- Redux Toolkit & React JS
- Next JS
- TypeScript & JavaScript
- FastAPI, Node.js, & NestJS
- Python & Java Spring Boot
- Generative AI & Agentic AI
- Google Gemini & Vertex AI
- Dialogflow CX & DSPy
- RAG, NL2SQL
- Prompt Engineering
- Slack Bot & AI Copilot Dev
- GCP (Cloud Run, Cloud Job)
- GCP (BigQuery, Cloud SQL)
- Docker, GitLab CI/CD & Jenkins
- PostgreSQL, Redis & Vector DB
- Microservices & Microfrontends
- SSO, RBAC & API Security

Results-driven System Architect Engineer with 8+ years of professional experience, including **5+ years in Software Development, Cloud Technologies, and Enterprise Application Architecture**, along with 4 years in Mechanical Design and Production Planning. Experienced in building **scalable web applications and cloud-native solutions** using **ReactJS, Next.js, TypeScript, Node.js, NestJS, Python, PostgreSQL, and Google Cloud Platform (GCP)**.

Specialized in **Generative AI, Agentic AI, Conversational AI, and Intelligent Automation** using **Google Gemini, Vertex AI, Dialogflow, DSPy, BigQuery, Cloud Run, and Cloud SQL**. Hands-on experience in developing **AI chatbots, Slack bots, AI copilots, NL2SQL platforms, RAG solutions, classification engines, and workflow automation systems**. Strong expertise in cloud architecture, microservices, API development, CI/CD pipelines, and end-to-end solution delivery, with a proven track record of driving innovation through AI-powered enterprise solutions.

WORK EXPERIENCE

Engr III Cslt-Sys Architect

Mar 2024 - Present

Verizon Data Services India Pvt Ltd

AI, Agentic AI & Enterprise Automation

Designed and delivered enterprise-grade AI platforms, Agentic AI solutions, conversational assistants, and intelligent automation systems leveraging Google Gemini and Vertex AI across multiple Verizon initiatives.

- Architected and delivered multiple enterprise-grade AI solutions using **Google Gemini, Vertex AI, Dialogflow CX, DSPy, Cloud Run, Cloud Jobs, BigQuery, Cloud SQL**, Redis, and Slack integrations.
- Designed and deployed Agentic AI frameworks with workflow orchestration, reasoning layers, Assist integration, model fine-tuning, memory management, and enterprise deployment pipelines.
- Led development of Gemini Enterprise Slack Bots, Hypercare Support Bots, Citrix Support Bots, YubiKey Automation Bots, Value Assistant Bots, and internal AI copilots.
- Engineered **vector-based and vectorless RAG architectures**, enabling document retrieval, **conversational AI**, self-training workflows, and enterprise knowledge management.
- **Built zero-storage conversational architectures** utilizing browser-scoped memory and session context management to ensure privacy-compliant AI interactions.
- Developed classification APIs, reranking systems, context-understanding engines, and AI-driven routing frameworks for intelligent enterprise support automation.

Generative AI, Multi-Modal AI & Knowledge Systems

Designed and implemented enterprise Generative AI solutions enabling intelligent search, content generation, document understanding, video intelligence, and natural language interaction across multiple business applications.

- Developed **NL2SQL solutions** leveraging Gemini and **BigQuery**, enabling users to generate and execute database queries using natural language.
- Engineered **vector-based and vectorless retrieval systems** supporting semantic search, knowledge discovery, and conversational AI experiences.

TECHNICAL SKILLS

- Single SPA & MFE
- Jest & React Testing Library
- AWS Lambda & DynamoDB
- MS SQL & PostgreSQL
- HTML5, SCSS & CSS3

TOOLS & PLATFORMS

- VS code
- IntelliJ
- Jupyter Notebook
- Postman & Thunder Client
- Git, GitHub & Bitbucket
- OneJIRA & Jenkins
- PuTTY
- Google Cloud Workbench
- Google Colab
- Gemini Enterprise
- Google Cloud Connectors
- Casino (SSO Platform)

SOFTWARES

- Power BI
- PTC Creo Parametric 8.0.10
- Adobe Illustrator
- Creo Illustrate 6.0
- Adobe Photoshop
- Teamcenter Visualization
- Arbortext XML Editor
- Windchill PLM

- Built **multi-modal AI applications** supporting text, document, image, audio, and video processing workflows.
- Developed video intelligence pipelines involving video upload, audio extraction, transcription, embedding generation, retrieval, and conversational querying capabilities.
- Implemented **OCR, similarity search, text embedding generation**, reranking, and document understanding services using Vertex AI and custom AI frameworks.
- Designed self-learning and context-aware AI workflows to improve response quality, retrieval accuracy, and user experience across enterprise platforms.

Data Science, Analytics & AI Insights

Designed and developed AI-powered analytics platforms to measure employee experience, application health, collaboration effectiveness, and operational performance.

- Architected APIs for Employee Experience Scores (eQES), App Scores, GMeet Metrics, Friction Scores, and Telemetry Analytics.
- Developed **scoring frameworks and AI-powered reasoning engines** to generate insights, summaries, and recommendations.
- Built **Cloud Jobs and data pipelines** for data collection, processing, and analytics reporting.
- Developed **HR Insight platforms with conversational AI**, dashboarding, and Google Sheets integration.
- Implemented data validation, metric calculations, and performance analytics to improve reporting accuracy.

Enterprise Application Development & Architecture

Designed and delivered scalable enterprise applications, internal platforms, and cloud-native solutions supporting AI adoption, asset management, user engagement, and business automation.

- Developed **full-stack applications** using ReactJS, Next.js, TypeScript, NestJS, FastAPI, Python, PostgreSQL, and Cloud SQL.
- Built conversational AI platforms, admin dashboards, document indexing systems, prompt management solutions, and enterprise workflow applications.
- Implemented **SSO, RBAC, JWT authentication, API security**, caching strategies, and microfrontend integrations across multiple applications.
- Delivered platforms including **VZGPT, InfoGenie, Prompt Gallery, VZArena, IT Asset Management, Return Asset Management, and VZGenie**.
- Established CI/CD pipelines, deployment automation, reusable frameworks, and cloud-native architectures on GCP.

Freelance Full Stack Developer

Jun 2024 - Present

LIVING FIRE PTY LTD, Victoria, Australia

- Designed, developed, and maintained a scalable SEO-optimized e-commerce platform using Next.js, PostgreSQL, TanStack Query, and Server Actions.
- Architected application workflows and integrated AWS S3 for secure media storage, retrieval, and application scalability.
- Developed a custom CMS and Admin Portal enabling business users to manage products, categories, inventory, content updates, and website operations without technical assistance.
- Implemented advanced SEO strategies including metadata optimization, sitemap generation, Open Graph integration, semantic HTML, and performance optimization.
- Developed user-focused features including authentication, product management, shipment tracking, payment gateway integration, and responsive user experiences.
- Managed production deployments on Vercel, provided ongoing maintenance and bug fixes, and collaborated directly with Australian client stakeholders to deliver continuous platform enhancements and support.

CORE SKILLS

- Solution Architecture
- Technical Leadership
- Problem Solving
- Critical Thinking
- System Design
- Stakeholder Management
- Cross-Functional Collaboration
- Mentoring & Knowledge Sharing
- Agile Delivery
- Time Management

LANGUAGES

- English
- Tamil

Specialist - Software Engineering

Feb 2022 - Feb 2024

LTIMINDTREE PVT LTD

L'Oréal DDXL Loyalty - Project 1

Developed a scalable Loyalty Management Platform integrated with CRM systems, leveraging a centralized rule engine to process customer rewards, points calculations, and loyalty program workflows.

- Developed and maintained the DDXL platform using React, Redux Toolkit, Node.js, DynamoDB, and AWS OpenSearch.
- Delivered scalable solutions and resolved complex technical challenges across 70+ integrated modules.
- Designed RESTful APIs and analytics services supporting KPI reporting and dashboard visualizations.
- Improved code quality through unit testing, performance optimization, reusable component architecture, and SonarQube best practices.
- Provided production support, bug resolution, and end-to-end testing across frontend and backend systems.

PHILIPS SAAS-Factory Narmada - Project 2

SaaS Frontdoor is an enterprise application launchpad that provides users with a centralized portal to access authorized applications through a unified interface, enabling seamless navigation and Single Sign-On (SSO) across integrated systems.

- Developed scalable microfrontend applications using Single-SPA, integrating React and Angular-based modules.
- Designed shared state management and communication mechanisms across MFEs using Redux architecture.
- Implemented platform-wide capabilities including internationalization (i18n), feature management, geolocation, and common integration services.
- Supported production releases through bug fixes, manual E2E testing, and technical documentation.
- Developed the **intersect API call** and **geolocation** call in the root config Application.

CISCO System Inc - Project 3

Developed the Cisco Integrated Management Controller (CIMC), an enterprise web application used to monitor, configure, and manage Cisco UCS servers through an embedded Baseboard Management Controller, enabling efficient server administration and lifecycle management.

- Developed reusable UI components and test cases using React Storybook.
- Built custom React hooks to improve component reusability, state management, and application performance.
- Designed and implemented REST APIs using Spring Boot and Spring MVC.
- Integrated APIs using React Query and developed complex business logic for optimized application workflows and rendering.

Front End Developer (Service Information Developer)

Jan 2021 - Feb 2022

Caterpillar Inc, Chennai (Randstad Payroll)

CAT SIS 2.0 Web - Project

- Developed the Add To Cart functionality on SIS2.0 Website using React JS, JavaScript, HTML5, and Bootstrap.
- Enhanced the user interface, Mobile Responsiveness, and **Payment Form** for the SIS2.0 Website using **React Hook Form** and **Stripe API** with Material UI.
- Fetching the data based on Sales models and Prefixes to display the parts information and its respective manuals.

Caterpillar Inc, Chennai (Randstad Payroll)

- Service Information Types: Operation and Maintenance Manuals Specifications
- Manual I Service Magazine I Technical Information Bulletin I Special Instruction I
- Visual Service Procedure I Technical Analysis I Testing, and Adjusting.
- Primary for developing the Work Tool (Grapple, Bucket, Hammer, etc.) And Paving products (Pavers, Cold Planers, and Compactors) Operation and Maintenance Manuals (OMM) as per the NPI inputs and PRWB (BOM) Design changes.
- Analyzed the input from 2D engineering drawings with GD&T knowledge for authoring and creating the illustration effectively.
- Developing the Macro sheets using VBA coding for the research process, Comparison of MIST, and NPI Project Research Sheet, and also coordinating with other authors to create the new macros depends on feasibility.

Production Planning Control Engineer**Aug 2016- Aug 2018****Kunaerospace Pvt Ltd, Chennai**

- Converting the customer schedules into the production plan under priority mode as well as a FIFO.
- Responsible for making monthly turnover budget by billing Material and maintaining it.

PERSONAL AI PROJECTS**AI OCR Platform – Founder & Developer**

- Designed and developed a production-ready AI-powered OCR platform for extracting and structuring information from images and documents.
- Built using Next.js, Python, PostgreSQL, Supabase, AI/OCR services, Vercel, and Railway.
- Currently deployed and actively maintained as a live SaaS application

AI Job Application Automation Platform – Founder & Developer

- Developed an AI-powered platform for resume optimization, job matching, and automated job application workflows.
- Built using Next.js, Node.js, PostgreSQL, Supabase, AI APIs, Vercel, and Railway.
- Currently deployed and actively maintained as a live web application.

EDUCATION**Bachelor of Engineering: Mechanical Engineering****2012- 2016**

Sri Ramanujar Engineering College - Chennai

GPA: 7.4

CERTIFICATIONS

- Google AI Essentials – Google
- Google AI Fundamentals – Google
- Google AI Professional Certificate – Google
- Google Cloud Skills Boost Badges (Multiple) – Google Cloud
- Google Cloud Hero Program – Google Cloud
- MERN Stack Front To Back: Full Stack React, Redux & Node.js – Udemy
- PTC Creo Parametric (Design & Simulation) – Udemy
- CATIA V5 (Design & Assembly) – Udemy